



Smarter Google Photos

Google Photos now manages space smartly by allowing you to bulk delete backed up photos
<http://dgit.in/GPBlkDlt>



Tumblr GIF on Phones

After various others, Tumblr finally launches GIF maker app for smartphones
<http://dgit.in/TumblrGIF>



Every month we recognise the best article written by our community members and publish it here. Keep up the good work, Digitians!

Turn your regular home into a smart home!

With 2016 looming on the horizon, what could be a better way to welcome the new year than with a DIY project that saves your electricity bills and breathes new life into your appliances?



Praneeet Sah

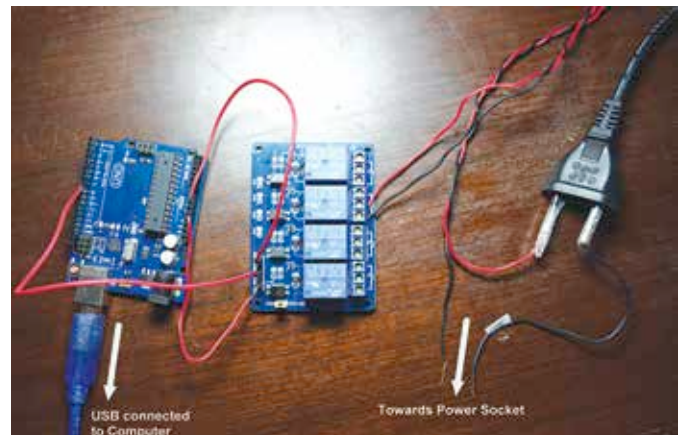
The idea of controlling your electric appliances from as far as the other end of the earth might sound like a sci-fi plot device straight out of an Isaac Asimov story. But in reality, it's quite feasible to turn everything – right from your washing machine, toaster, television, and air conditioners to even the lights and fans in your room – on and off from afar,

provided there's an Internet connection on both ends. All you need is a device that is capable of opening a website. So your phone, laptop, tablet or even smart-watch can do the magic!

This whole project is aimed at achieving the aforementioned task at the lowest possible cost by using duplicate-yet-good components and by substituting elements such as wires and Ethernet shields with “jugaad”.

The main component here is an Arduino and a 5-Volt Relay (go through ‘FastTrack: Arduino for Everyone’ if you're not familiar with it). In a nutshell, Arduino can be described as a microcontroller that acts just like a computer but on a very small scale in terms of power and features. It is palm-sized and can cost anywhere between ₹500 and 2000 (Arduino is actually an open-source product which is why the duplicate-yet-workable model goes for ₹500 while the original one goes for the higher limit). In this project, I've used an Arduino ‘Uno’ because it gets the work done (and is more famous) and has got the option to add more sensors given the greater number of I/O pins.

The ‘5 Volt Relay’ is basically a switch that can be controlled through digital signals (i.e via computer). It comes in different ‘channels’ which refers to the number of switches or devices you can control with it. Channels 1, 2, 4, and 8 are the most common types and I've used the “4 Channel 5 Volt Relay” but you're free to use any other relay. The 4-Channel relay will have 4 individual switches, each equipped with 3 connecting ports – Common, Normally Closed (NC) and Normally Open (NO). Don't make the rookie mistake of confusing these for the traditional positive, negative and the earthing ports that



Overview of the whole circuit

are present in an electric switch that you'd generally find on switch boards.

Since the objective is to control appliances from anywhere, we would, first and foremost, require a website. So the first step is to acquire a domain name and hosting space. You could either buy them or get them for free. If you are just experimenting, I'd recommend setting up your own local server using XAMPP, (a software which is absolutely free. It's a cross-platform web server that you can setup within minutes and you can then drop your files into the htdocs folder within the XAMPP installation. Accessing those files is as simple as typing 127.0.0.1 in your web browser. You will need the file to be named as index.php or index.html if you wish to access the site without having to manually type in the entire file name of the html / php file. Read up about it to know more, for you need to have created at least a ‘hello world’ website in the past to proceed with this venture and it's one of the easiest steps in this project.

In addition, you'd require some jumper wires; however, I wasn't able to find them anywhere in the cities I reside in – Bareilly and Varanasi. So you could either order them online or